

Week 1 — Dataset Inspection Report

1. Source and Dimensions

Dataset: Telco Customer Churn (Kaggle / IBM Sample Data Sets — <https://www.kaggle.com/datasets/blatchar/telco-customer-churn>).

Dimensions: 7,043 rows × 21 columns → 7,043 unique customers. No duplicate customerID, no fully duplicated rows.

2. Data Types

Type	Columns
int64	SeniorCitizen, tenure
float64	MonthlyCharges
object (text)	customerID, gender, Partner, Dependents, PhoneService, MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies, Contract, PaperlessBilling, PaymentMethod, TotalCharges, Churn

Important point: TotalCharges is stored as text (object) whereas it clearly represents numerical amounts — a classic pitfall of this dataset to correct before any processing.

3. Missing Values

A standard check (`isna().sum()`) returns 0 on all columns. However, this does not mean the data is perfectly clean: since TotalCharges is stored as text, hidden missing values in the form of empty strings (" ") are not detected by this standard check.

Upon further investigation, 11 rows present an empty string in TotalCharges instead of a number. They all share one commonality: tenure = 0, meaning these are customers who have just signed up (less than one month of tenure), and thus have not yet accumulated a total bill. This is not a random data entry error, but business consistency (0 months × monthly charge = nothing to accumulate).

Recommendation: Do not ignore these 11 rows. Convert TotalCharges to numerical then replace the missing values with 0 (equivalent to tenure × MonthlyCharges in this specific case, since tenure = 0).

4. Duplicates

- Fully duplicated rows: 0
- Duplicated customerIDs: 0

Each row represents a unique customer — no duplicate cleaning is necessary.

5. Distribution of the Target Variable (Churn)

Class	Number of customers	Percentage
No	5,174	73.5%
Yes	1,869	26.5%

This is a moderate class imbalance (about 1 to 3). This confirms that accuracy alone will not be sufficient as an evaluation metric: recall, precision, F1-score, and ROC-AUC should be prioritized (see the Business Understanding Report).

6. Input Variables (features) — Answer to Question 3

Once customerID (unique identifier with no predictive value) and Churn (the target) are removed, there are 19 usable input variables left, split into two groups:

- **Numerical variables (3):** tenure, MonthlyCharges, TotalCharges.
- **Categorical variables (16):** gender, SeniorCitizen, Partner, Dependents, PhoneService, MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies, Contract, PaperlessBilling, PaymentMethod.

The details of the required preprocessing for these variables (encoding, scaling, handling class imbalance) are presented in the Machine Learning Proposal.

7. Summary

- 7,043 customers, 21 columns, no "classic" missing values, no duplicates.
- TotalCharges incorrectly typed (text) with 11 hidden blanks, all linked to tenure = 0 — corrected by numerical conversion + imputation with 0.
- Imbalanced target (73.5% / 26.5%) → direct impact on the choice of evaluation metrics.

- 19 input variables identified: 3 numerical, 16 categorical.